UFA Throwing EDA Project

CMSACamp 2026

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Analytical insights from ultimate frisbee remains limited

- A growing but still niche sport → insights are limited
- Explore how throw patterns reveal useful information about teams' strategies
- Data from the **Ultimate Frisbee Association (UFA)** (4 seasons, 2021-2024)

Ian Piper,
Pittsburgh Thunderbirds

Each row in the original dataset represents a specific throw in a match

Throw ID	Thrower X	Thrower Y	Distance (yard)	Time Remaining (s)	Goal
1	1.02	15.14	12.43	2,880.00	0
2	–8.17	23.51	10.95	2,876.58	0
3	2.18	27.08	13.98	2,873.16	0
4	–10.21	33.56	9.23	2,869.74	0
5	–3.94	26.79	19.37	2,866.32	0
6	12.96	36.26	11.37	2,862.89	0

…alongside game, team, and thrower information.

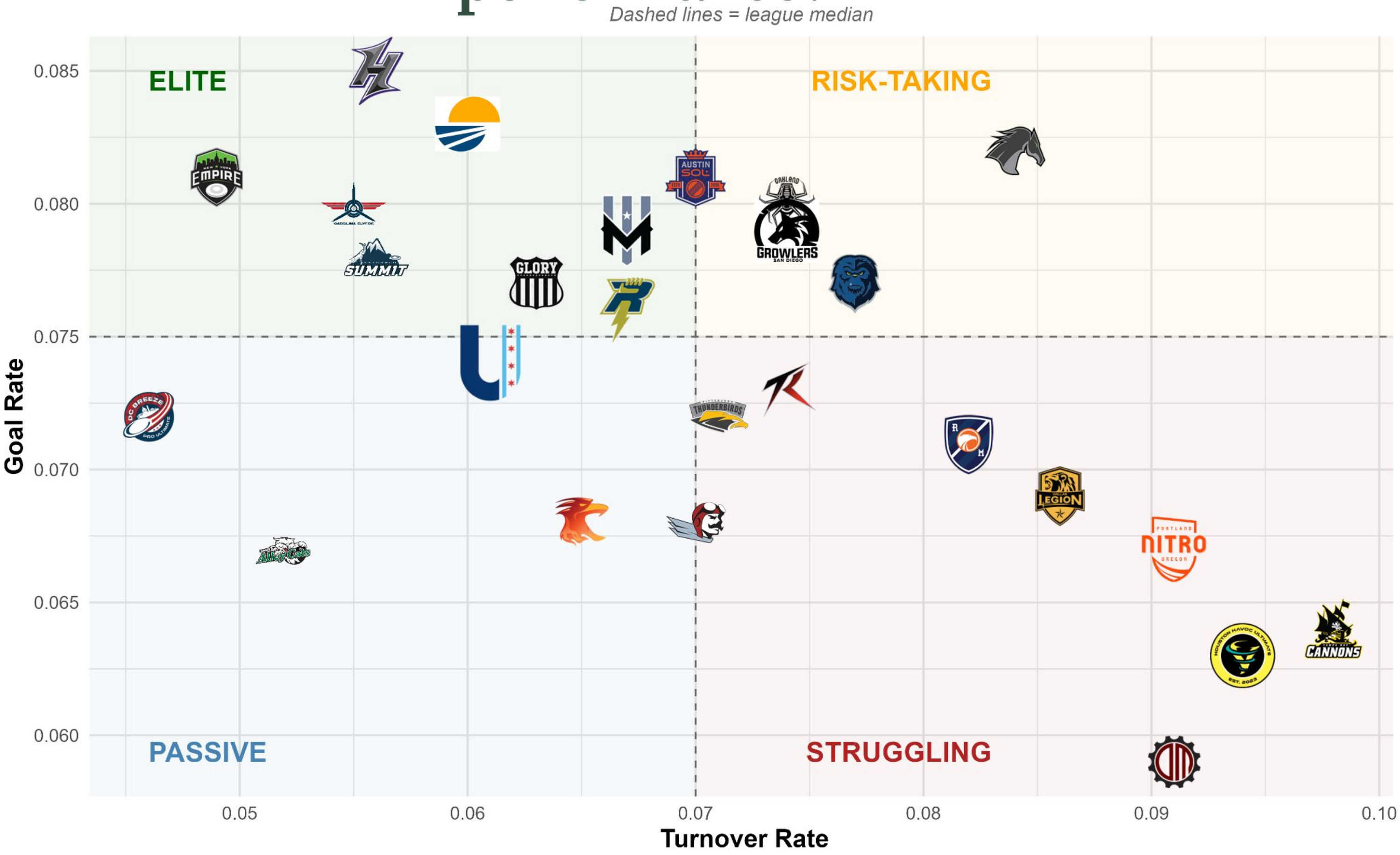
Team-level data is aggregated across all games mentioned in the dataset

Team	Win (%)	Goal Rate (%)	Turnover Rate (%)	Goals Scored	Goals Conceded	Goal Difference
Alleycats	0.460	0.067	0.052	984	991	-7
Aviators	0.413	0.068	0.070	809	881	-72
Breeze	0.800	0.072	0.046	1045	835	210
Cannons	0.125	0.064	0.098	368	539	-171
Cascades	0.408	0.077	0.077	967	989	-22
Empire	0.849	0.081	0.049	1121	855	266

#1

How can we evaluate the current performance of a team?

How can goal and turnover rates indicate team performance?



What matters more: scoring more or making fewer mistakes?

UFA Team Efficiency Table

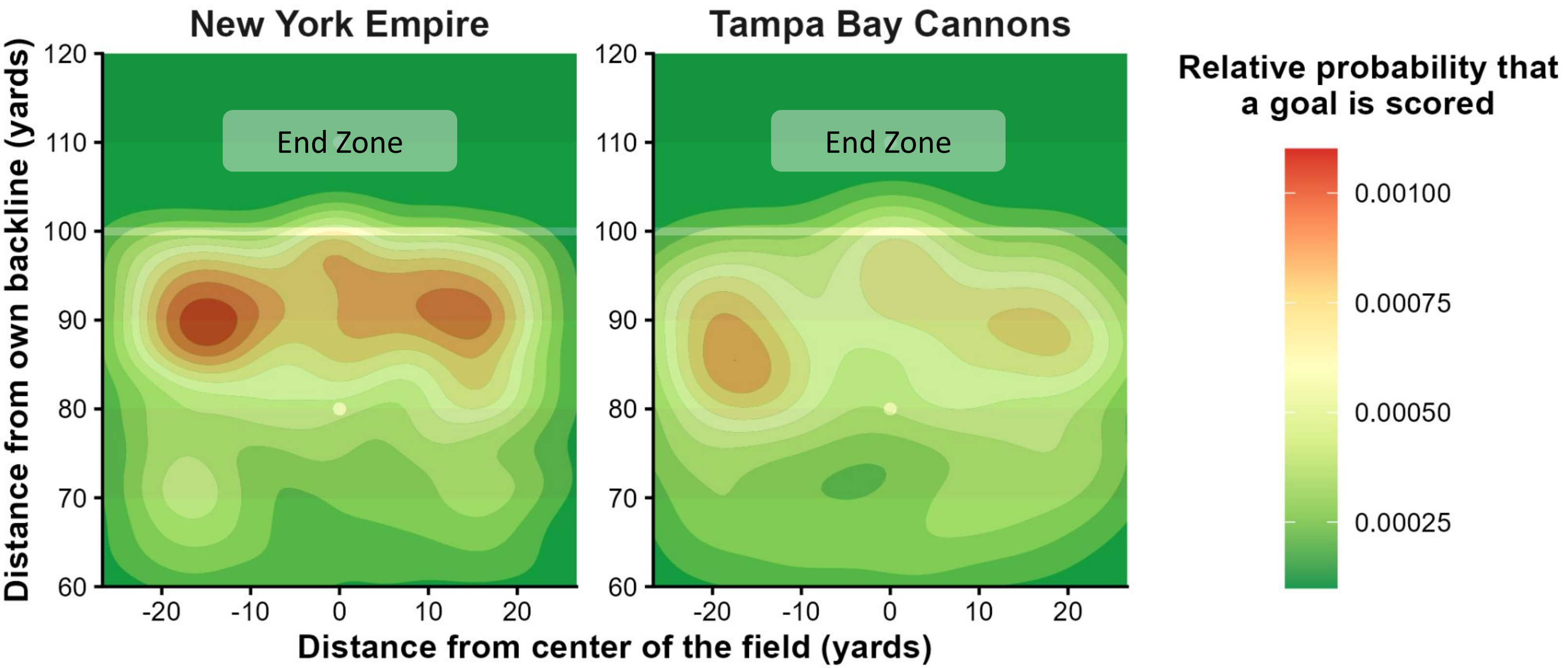
Grouped by performance quadrant

Team	Win %	Goal Rate	Turnover Rate	Goal Difference
Elite				
empire	84.9%	8.1%	4.9%	266
shred	80.0%	8.3%	6.0%	139
windchill	78.2%	7.9%	6.7%	191
flyers	77.4%	8.0%	5.5%	194
hustle	69.6%	8.5%	5.6%	168
summit	65.8%	7.8%	5.6%	128
sol	59.2%	8.1%	7.0%	104
radicals	55.1%	7.6%	6.7%	72
glory	50.0%	7.7%	6.3%	19
Passive				
breeze	80.0%	7.2%	4.6%	210
union	67.9%	7.4%	6.1%	140
alleycats	46.0%	6.7%	5.2%	−7
aviators	41.3%	6.8%	7.0%	−72
phoenix	41.3%	6.8%	6.5%	−56
Risk-Taking				
growlers	54.0%	7.9%	7.4%	3
spiders	45.5%	8.0%	7.4%	8
cascades	40.8%	7.7%	7.7%	−22
outlaws	22.2%	8.2%	8.4%	−51
Struggling				
havoc	34.8%	6.3%	9.4%	−99
thunderbirds	33.3%	7.2%	7.1%	−57
royal	32.6%	7.1%	8.2%	−118
rush	27.3%	7.3%	7.4%	−139
legion	20.0%	6.9%	8.6%	−184
cannons	12.5%	6.4%	9.8%	−171
nitro	11.4%	6.7%	9.1%	−239
mechanix	2.1%	5.9%	9.1%	−427

#2

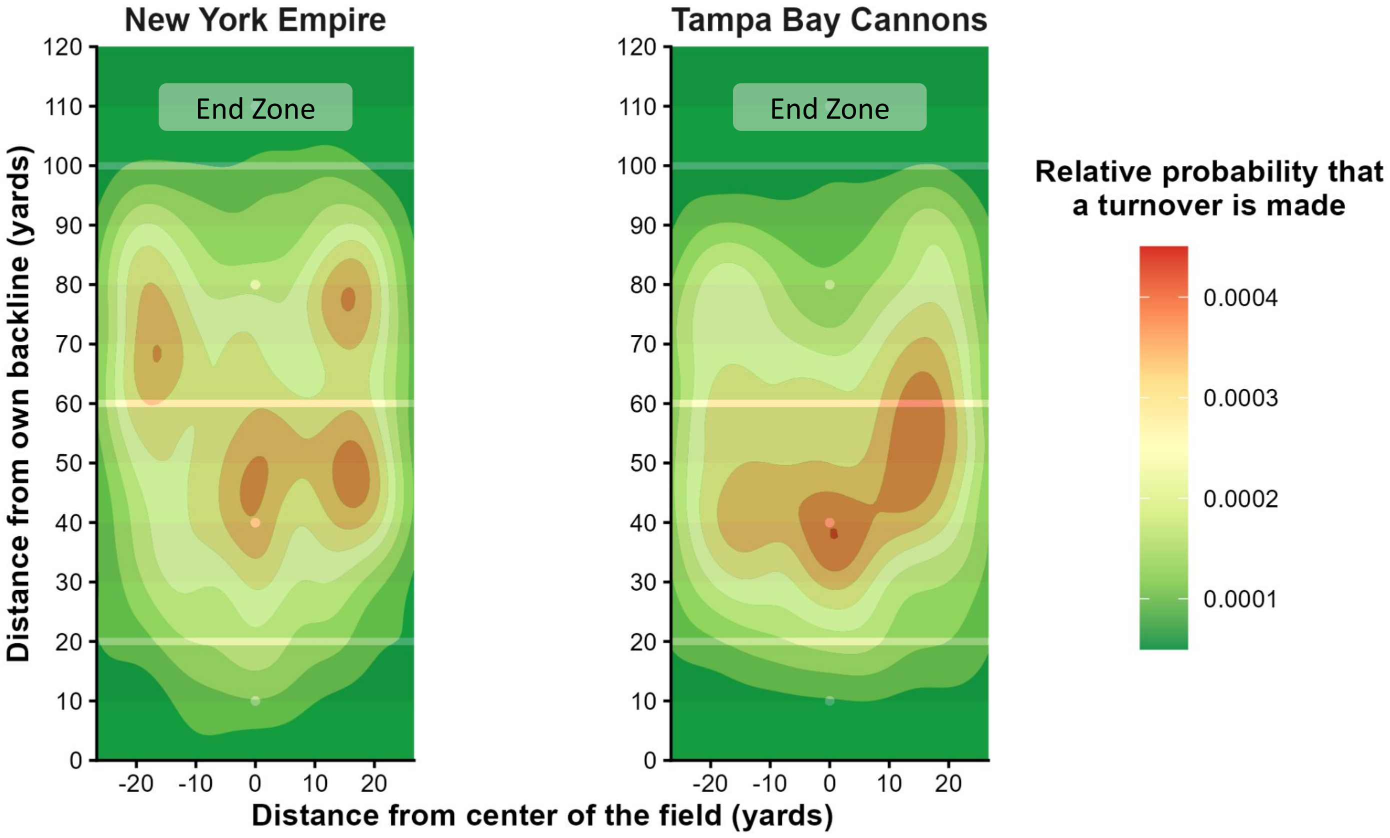
If two teams have the same strategy, what metric separates high-performing and low-performing teams?

Both the Empire and Cannons have (approximately) the same scoring pattern …



Data courtesy of Ultimate Frisbee Association & Eberhard, Miller, and Sandholtz (2025).

…but they have different turnover patterns

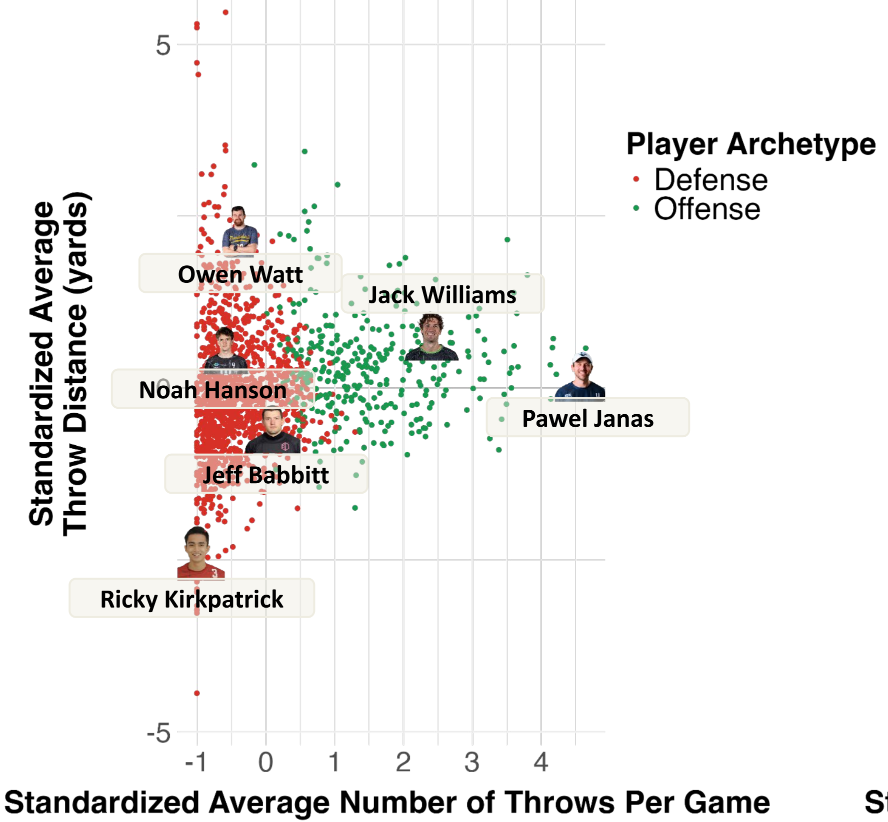


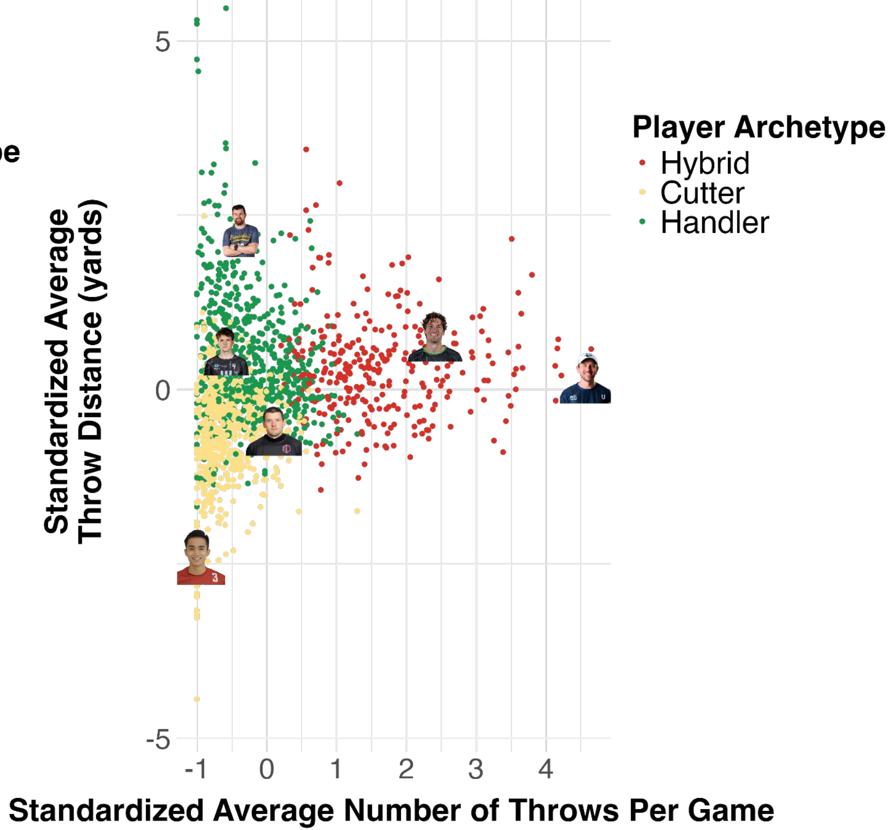
Data courtesy of Ultimate Frisbee Association & Eberhard, Miller, and Sandholtz (2025).

#3

How do we characterize different positions/playing styles within an ultimate frisbee team?

While it is easier to distinguish between defensive and offensive players, a player's specific role is not clear-cut





To sum up …

- **Goal and turnover rates** well reflect team success in the UFA.
- Among teams with **similar scoring patterns**, elite teams exhibit more **spatially diverse turnover locations**, suggesting a wider range of attacking options.
- UFA players frequently switch between **handler and cutter roles**, though some show clear **specialization in defensive play**.

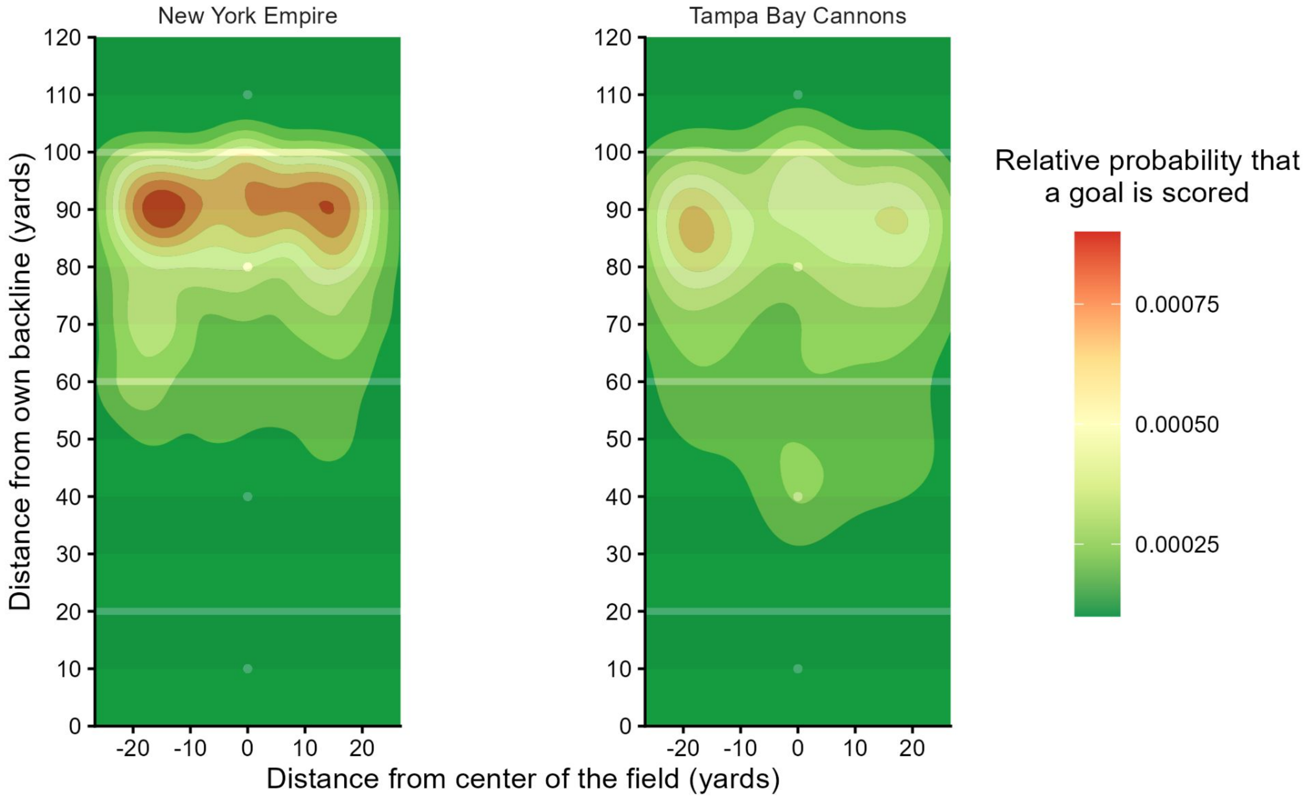
Future work:

- Incorporate defensive metrics (e.g., interceptions per game) to better characterize team and player styles.
- Track disc trajectories throughout possessions to quantify attacking strategies and patterns.

Kyle Henke,
Austin Sol

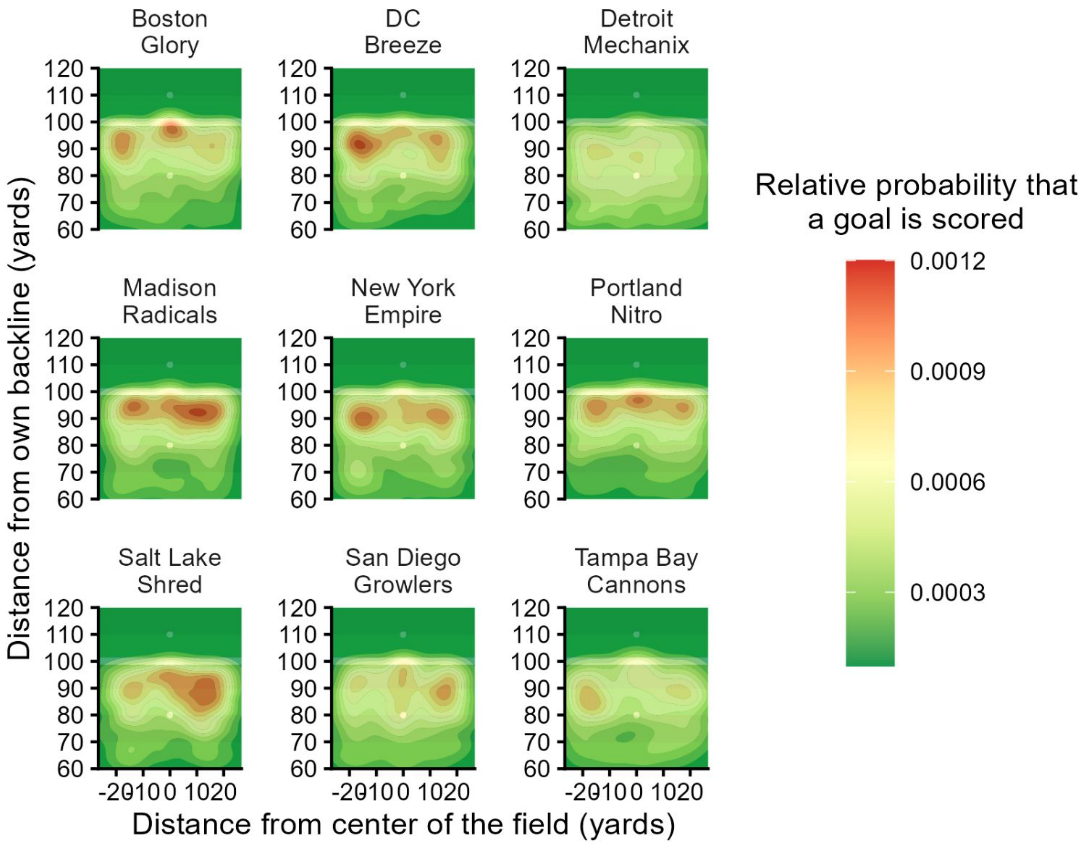
Appendix

1. 1 Scoring patterns of the Empire and the Cannons on the entire field

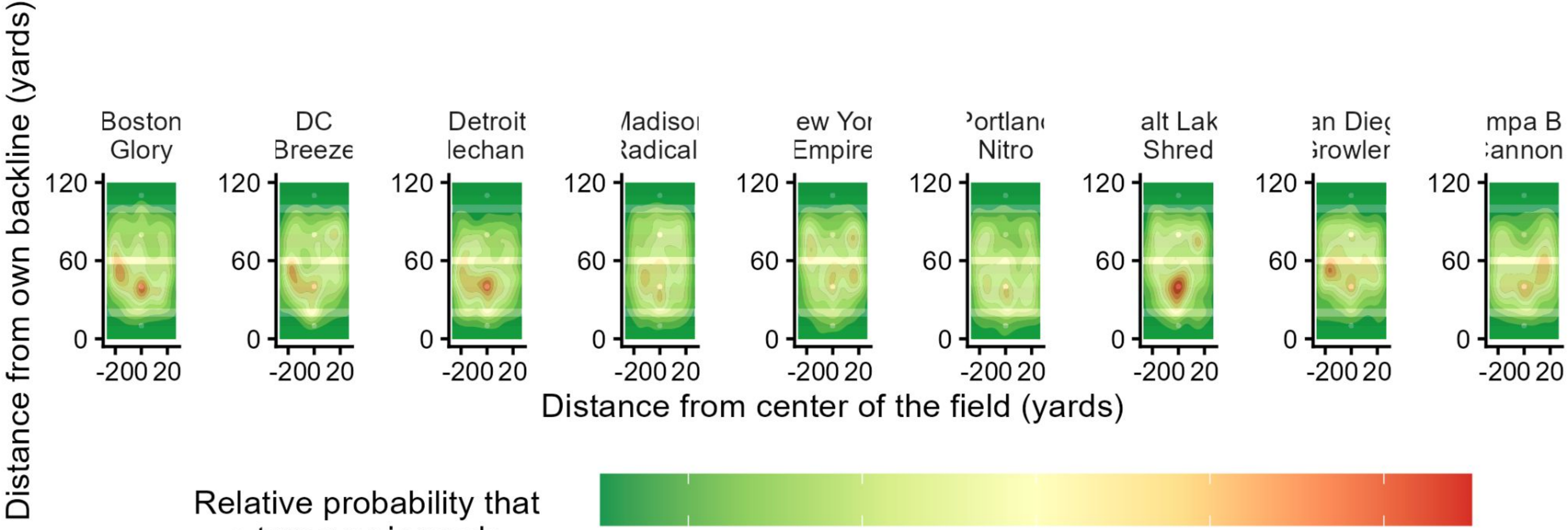


Data courtesy of Ultimate Frisbee Association & Eberhard, Miller, and Sandholtz (2025).

1.2. Scoring patterns of the top 3, “middle 3”, and bottom 3 teams in the dataset



1.3. Turnover patterns of the top 3, “middle 3”, and bottom 3 teams in the dataset



Data courtesy of Ultimate Frisbee Association & Eberhard, Miller, and Sandholtz (2025).